Algebraic expressions

Whole expressions, fractional expressions, and the values of the variable that are simply not allowed.

LESSON 4–5 | 2 × 40 MIN | ALGEBRA 8, §1 | STAGE 9 · 2.1–2.2

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Tell a whole (integer) expression from a fractional one.
- Find the permissible values of the variable — the domain of an expression.
- Evaluate an algebraic expression for given values of its letters.
- Write a described quantity as an algebraic expression.

TEXTBOOK

National	§1, pp. 7–11
Cambridge	Learner’s Book pp. 22–29
Workbook	Workbook 2.1–2.2

01 Explanation

Two kinds of expression

An expression built from numbers and letters using addition, subtraction, multiplication and raising to a whole power — but **never dividing by a letter** — is called a **whole** or **integer expression**.

$3x^2 - 5x + 1$ $\quad \frac{2a}{7} \quad$ $(x - y)^3$

If the expression divides by something containing a letter, it is a **fractional expression**.

$\frac{5}{x} \quad \frac{x+1}{x-3} \quad 2 + \frac{1}{a}$

Note the difference between $\frac{2a}{7}$ and $\frac{7}{2a}$. The first divides by the *number* 7 and is whole; the second divides by $2a$ and is fractional. It is the letter in the denominator that changes everything.

Permissible values of the variable

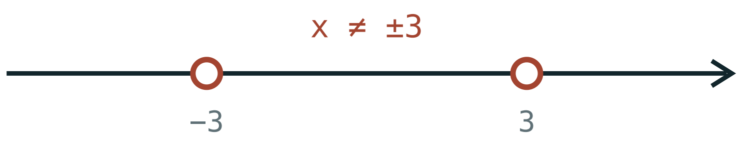
A whole expression has a value for *every* value of its letters. A fractional expression does not: division by zero has no meaning.

> DEFINITION
>
> The **permissible values** of the variable are all the values for which the expression has a value — that is, all values that do not make any denominator zero.

So for $\frac{x+1}{x-3}$ we must exclude $x = 3$, and we write $x \neq 3$.

To find them: set every denominator equal to zero, solve, and exclude those answers.

$$x^2 - 9 = 0 \Longrightarrow x = 3 \text{ or } x = -3, \text{ so } x \neq \pm 3$$



The two excluded points are drawn as open circles — the line has holes there.

> CAREFUL
>
> Some denominators never vanish. $x^2 + 4$ is at least 4 for every real x, so $\frac{3}{x^2+4}$ is defined for **all** values of x.

The value of an expression

To evaluate, substitute and calculate — but check the value is permissible first.

$\frac{a^2 - b^2}{a - b}$ at $a = 7, b = 3$: the denominator is $7 - 3 = 4 \neq 0$, so the value is $\frac{49 - 9}{4} = 10$.

At $a = 3, b = 3$ the denominator is 0 — the expression has no value there at all.

Writing expressions from words

- “a number 5 more than three times x” → $3x + 5$
- “the reciprocal of $x - 5$” → $\frac{1}{x - 5}$, with $x \neq 5$
- “the average speed for s km covered in t hours” → $\frac{s}{t}$, with $t \neq 0$

02 Worked examples

EXAMPLE 1 ***Find the permissible values of*** $\mathbf{\frac{x + 1}{x^2 - 5x + 6}}$

1. $x^2 - 5x + 6 = 0$

   Set the denominator to zero.

2. $(x - 2)(x - 3) = 0$

   Factorise.

3. $x = 2$ or $x = 3$

   These are the forbidden values.

4. $x \neq 2, x \neq 3$

   Everything else is permitted.

ANSWER

$x \neq 2$ and $x \neq 3$

EXAMPLE 2 ***For which x is*** $\dfrac{\mathbf{x-2}}{\mathbf{x+3}}$ ***equal to zero?***

1. A fraction is zero exactly when its **numerator** is zero and its denominator is not.
   Both conditions matter.

2. $x - 2 = 0 \Longrightarrow x = 2$
   Numerator condition.

3. *at* $x = 2$ *the denominator is* $5 \neq 0$ ✓
   Check the value is permissible.

ANSWER

$x = 2$

03 Interactive model

Slide x across the excluded value and let the class watch the fraction break.

Try a value of x

$$\frac{x + 1}{x^2 - 4}$$

x **3** | denominator **5** | value **0.8**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Algebraic expression	Algebraik ifoda	Алгебраическое выражение
Whole (integer) expression	Butun ifoda	Целое выражение
Fractional expression	Kasr ifoda	Дробное выражение
Variable	O‘zgaruvchi	Переменная
Permissible values	Mumkin bo‘lgan qiymatlar	Допустимые значения
Numerator	Surat	Числитель
Denominator	Maxraj	Знаменатель
Value of an expression	Ifodaning qiymati	Значение выражения
Undefined	Aniqlanmagan	Не определено
Reciprocal	Teskari son	Обратное число

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which of these is a **fractional** expression?

- $\frac{3x}{5}$
- $\frac{5}{3x}$
- $3x - 5$
- $(3x)^2$

The permissible values of $\frac{7}{x + 2}$ are:

- $x \neq 2$
- $x \neq -2$
- $x \neq 0$
- all values of x

For which x is $\frac{4}{x^2 + 1}$ undefined?

- $x = 1$
- $x = -1$
- $x = \pm 1$
- never — it is defined for all x

The value of $\frac{x^2 - 4}{x + 2}$ at $x = 3$ is:

- 1
- 5
- 0
- undefined

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Which of $3x/5$, $5/x$, $x/2 + 1$, $7/(x - 1)$ are fractional expressions?*

   ANSWER $5/x$ and $7/(x - 1)$

2. *Permissible values of* $\frac{1}{x}$

   ANSWER $x \neq 0$

3. *Permissible values of* $\frac{1}{x - 4}$

   ANSWER $x \neq 4$

4. *Permissible values of* $\frac{1}{x + 7}$

   ANSWER $x \neq -7$

5. *Value of* $\frac{x + 3}{x - 1}$ *when* $x = 5$

   ANSWER 2

6. *Value of* $\frac{2a}{a + 2}$ *when* $a = 4$

   ANSWER $\frac{4}{3}$

7. *Permissible values of* $\frac{3}{2x}$

   ANSWER $x \neq 0$

Medium · the standard question

7 PROBLEMS

1. *Permissible values of* $\dfrac{x+1}{x^2-9}$

ANSWER $x \neq 3, x \neq -3$

2. *Permissible values of* $\dfrac{2x}{x^2+1}$

ANSWER All values of x — the denominator is never 0.

3. *Permissible values of* $\dfrac{5}{x^2-4x}$

ANSWER $x \neq 0, x \neq 4$

4. *Value of* $\dfrac{a^2-b^2}{a-b}$ *when* $a = 7, b = 3$

ANSWER 10

5. *Value of* $\dfrac{x^2-4}{x+2}$ *when* $x = 3$

ANSWER 1

6. *Write the reciprocal of* $x - 5$ *and state its permissible values*

ANSWER $\dfrac{1}{x-5}$, $x \neq 5$

7. *For which* x *is* $\dfrac{x-2}{x+3} = 0$?

ANSWER $x = 2$

Hard · stretch and reason

7 PROBLEMS

1. *Permissible values of* $\dfrac{x - 1}{x^2 - 5x + 6}$

ANSWER $x \neq 2, x \neq 3$

2. *Permissible values of* $\dfrac{1}{x^2 + x}$

ANSWER $x \neq 0, x \neq -1$

3. *Solve* $\dfrac{x^2 - 9}{x - 3} = 0$

ANSWER $x = -3$ only — $x = 3$ is not permissible.

4. *Simplify, then evaluate* $\dfrac{x^2 - 6x + 9}{x - 3}$ *at* $x = 10$

ANSWER $x - 3 = 7$

5. *Permissible values of* $\dfrac{3}{|x| - 2}$

ANSWER $x \neq 2, x \neq -2$

6. *Permissible values of* $\dfrac{5}{x^2 - 7x + 10}$

ANSWER $x \neq 2, x \neq 5$

7. *For which x is* $\dfrac{x + 4}{x^2 + 4}$ *undefined?*

ANSWER For no value — $x^2 + 4 \geq 4$ always.

07 Homework

Homework — 5 problems

Algebra 8, §1, pp. 7–11. Cambridge Workbook 2.1 for the substitution practice.

1. *State the permissible values of* $\dfrac{x+5}{x^2-16}$
2. *State the permissible values of* $\dfrac{2}{x^2+3x}$
3. *Find the value of* $\dfrac{a+b}{a-b}$ *when* $a = 9$, $b = 4$
4. *For which* x *is* $\dfrac{x+6}{x-1}$ *equal to zero?*
5. A car covers s km in t hours. Write its average speed as an algebraic expression and state the permissible values of t.